Training-run guide

exp022 · 11 August 2026 · Draft

Contents

Abstract

Exp022 defines the collection's shared training runs and checkpoint bank. It specifies the motivation, parameterization, output-layer shape, and downstream consumers for six training-run types. Five run types comprise 87 trained cells spanning reference models and controlled sweeps over spike budget, inhibitory timescale, integration timestep, and recurrent initialization. TR-06 adds three PING cells trained across variable input rates with ten spiking output LIF neurons; each class logit is the corresponding neuron's output firing rate. Together, these runs provide the checkpoints used by the collection's training-dependent experiments.

Shared parameters

Unless a run-specific table says otherwise, every cell uses the following contract.

Parameter	Default	Meaning
Dataset	MNIST	784 normalized pixels encoded as independent Poisson channels
Presentation duration	200 ms	One static digit per training presentation
Integration timestep	0.1 ms	2,000 recurrent updates per presentation
Epochs	50	Training horizon for every production cell
Seeds	42, 43, 44	Three independently initialized cells per configuration
Minibatch	256	Presentations per optimization step
Optimizer learning rate	4×10^{-4}	Shared learning rate
Input population	784	One channel per image pixel
Excitatory population	1,024	Learned stimulus representation
Inhibitory population	256	PING feedback population; silent when E/I coupling is disabled
τ_{AMPA}	2 ms	Fixed excitatory synaptic decay
τ_{GABA}	6 ms	Default inhibitory decay at the gamma operating point
Input sparsity	0.95	Sparsity of the input projection
Surrogate slope	1	Spike-gradient surrogate parameter
Default readout	`mem-mean`	A **spiking** 1024 → 10 output-LIF layer. Its neurons emit spikes and reset, but classification logits are their membrane voltages averaged over time—not their spike counts
Stored projection shapes	784×1024; 1024×256; 256×1024; 1024×10	Input→E, E→I, I→E, and E→class, in source-to-destination orientation

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Training-run guide

TR-01 — Canonical full-data reference

Establish the highest-data, unconstrained COBA and PING reference. These checkpoints anchor accuracy and activity comparisons; they are not interchangeable with the 10%-MNIST no-budget cells. This training run is used by exp022.

Key parameter	Value	Why it differs
Architectures	COBA and PING	Provides a feedforward control and recurrent E/I model

Key parameter	Value	Why it differs
Training pool	70,000 samples	Uses all pooled MNIST rather than the sweep default of 7,000
Input rate	25 Hz maximum-pixel rate	Fixed-rate collection baseline
Spike budget	Off	Measures unconstrained capacity
Cells	2 architectures $\times$ 3 seeds $= 6$	Across-seed comparison for both models
Readout shape	1024 $\rightarrow$ 10 spiking LIF outputs	`mem-mean`: mean membrane voltage supplies the logits

Parameter rationale. Full data is the defining deviation. Keeping the spike budget off prevents regularization from confounding the architecture baseline.

TR-02 — Spike-budget sweep

Measure the accuracy–activity trade-off and test whether recurrent gamma gating preserves classification as the allowed spike count tightens. This training run is used by exp024, exp025, exp037, exp038.

Key parameter	Value	Why it differs
Architectures	COBA and PING	Direct architecture comparison
Training pool	7,000 samples	Keeps the 36-cell sweep tractable
Spike budget θ_u	off, 5, 2, 1, 0.5, 0.2 spikes/neuron/trial	Spans unconstrained through severe sparsity
Penalty strength	10^{-3} when enabled	Applies the upper-rate regularizer
Cells	2×6 settings $\times$ 3 seeds $=$ 36	Error bars at every frontier point
Readout shape	1024 $\rightarrow$ 10 spiking LIF outputs	`mem-mean`: mean membrane voltage supplies the logits

Parameter rationale. Only the spike cap and its penalty are swept. The smaller pool is a compute decision, so absolute rates must not be compared directly with the canonical full-data cells.

TR-03 — Inhibitory-timescale sweep

Separate task accuracy from the timescale of the E/I rhythm and measure how inhibitory decay controls gamma frequency. This training run is used by exp041, exp042, exp046.

Key parameter	Value	Why it differs
Architecture	PING	The manipulated quantity belongs to the recurrent inhibitory loop
Training pool	7,000 samples	Sweep-scale default
τ_{GABA}	4.5, 6, 9, 12, 18, 27 ms	Moves the inhibitory rhythm across a broad timescale range
Cells	6 settings $\times$ 3 seeds = 18	Across-seed estimate at each decay
Readout shape	1024 $\rightarrow$ 10 spiking LIF outputs	`mem-mean`: mean membrane voltage supplies the logits

Parameter rationale. τ_{GABA} is the only scientific variable. The 6 ms cell is the internal standard operating point.

TR-04 — Integration-timestep sweep

Test numerical stability at fixed physical presentation duration and expose the accuracy–compute trade-off of finer integration. This training run is used by exp044.

Key parameter	Value	Why it differs
Architecture	PING	Tests the recurrent reference model
Training pool	7,000 samples	Sweep-scale default
Δt	0.05, 0.1, 0.25, 0.5, 1 ms	Changes numerical resolution while holding 200 ms physical time fixed
Steps/presentation	4,000; 2,000; 800; 400; 200	Compute and activation memory scale inversely with Δt
Cells	5 settings $\times$ 3 seeds = 15	Across-seed stability check
Readout shape	1024 $\rightarrow$ 10 spiking LIF outputs	`mem-mean`: mean membrane voltage supplies the logits

Parameter rationale. Physical duration does not change. The 0.05 ms cells are the memory exception and require the large-memory Cambridge request.

TR-05 — Recurrent-initialization sweep

Determine whether the recurrent E/I regime is learned from generic initialization or must be built into the network before supervised training. This training run is used by exp049.

Key parameter	Value	Why it differs
Architecture	PING	Manipulates the recurrent loop directly
Training pool	7,000 samples	Sweep-scale default

Key parameter	Value	Why it differs
Loop conditions	frozen PING; trainable PING init; trainable zero init; trainable 0.1 init	Separates built-in dynamics from recurrence learned during task training
Trainable projections	W_{EI} and W_{IE} only in trainable conditions	The frozen condition is the mechanistic control
Cells	4 conditions × 3 seeds = 12	Across-seed comparison
Readout shape	1024 → 10 spiking LIF outputs	`mem-mean`: mean membrane voltage supplies the logits

Parameter rationale. Recurrent initialization and trainability move together by design. All feedforward and classifier settings remain on the PING recipe.

TR-06 — Variable-rate streaming bank

Train a PING classifier whose input distribution and output decision rule match variable-rate streaming inference. The resulting checkpoint bank supports inference across input rates and presentation durations. This training run is used by exp082.

Key parameter	Value	Why it differs
Architecture	PING	Target model for streaming inference
Training pool	7,000 samples	Uses the shared sweep-scale training set
Input-rate set	0.5, 1, 2, 5, 10, 25 Hz	Range selected by exp080
Sampling rule	Uniform categorical, independently per presentation	Makes rate variation part of the training distribution
Readout	`spike-rate`	Hidden E spikes drive ten spiking LIF class neurons; each logit is that class neuron's spike count divided by presentation duration in seconds
Readout shape	1024 → 10 spiking LIF outputs	Ten class neurons emit and reset throughout the presentation
Cells	1 recipe × 3 seeds = 3	Checkpoint bank expected by exp082

Parameter rationale. Uniform categorical sampling gives each input rate equal representation during training. The `spike-rate` reduction expresses every class logit in hertz, making the readout comparable across presentation durations. During streaming inference, digit boundaries reset the output LIF state while the hidden PING state remains continuous.

Results by training run

Each completed run reports one training-curve figure and one representative seed-42 raster. The figures summarize all configurations and seeds; the raster is a diagnostic example, not an across-seed statistic.

TR-01 results — Canonical full-data reference

Run status: complete · 6/6 cells represented

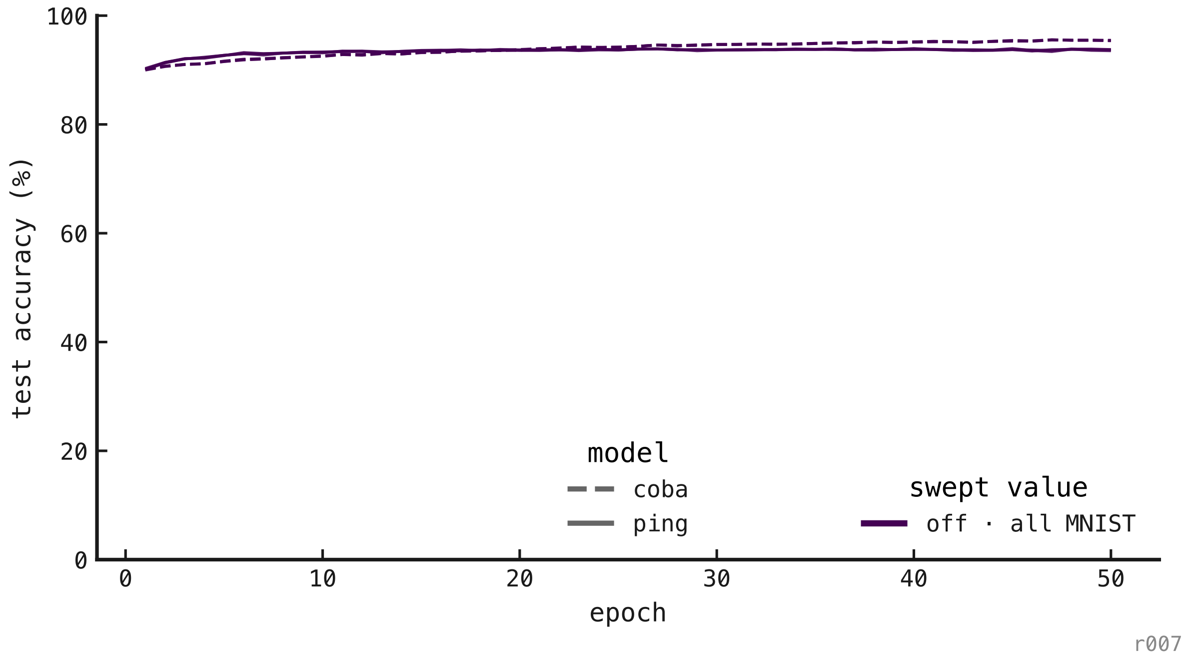

Figure 1: Training curves for the six full-data reference cells.

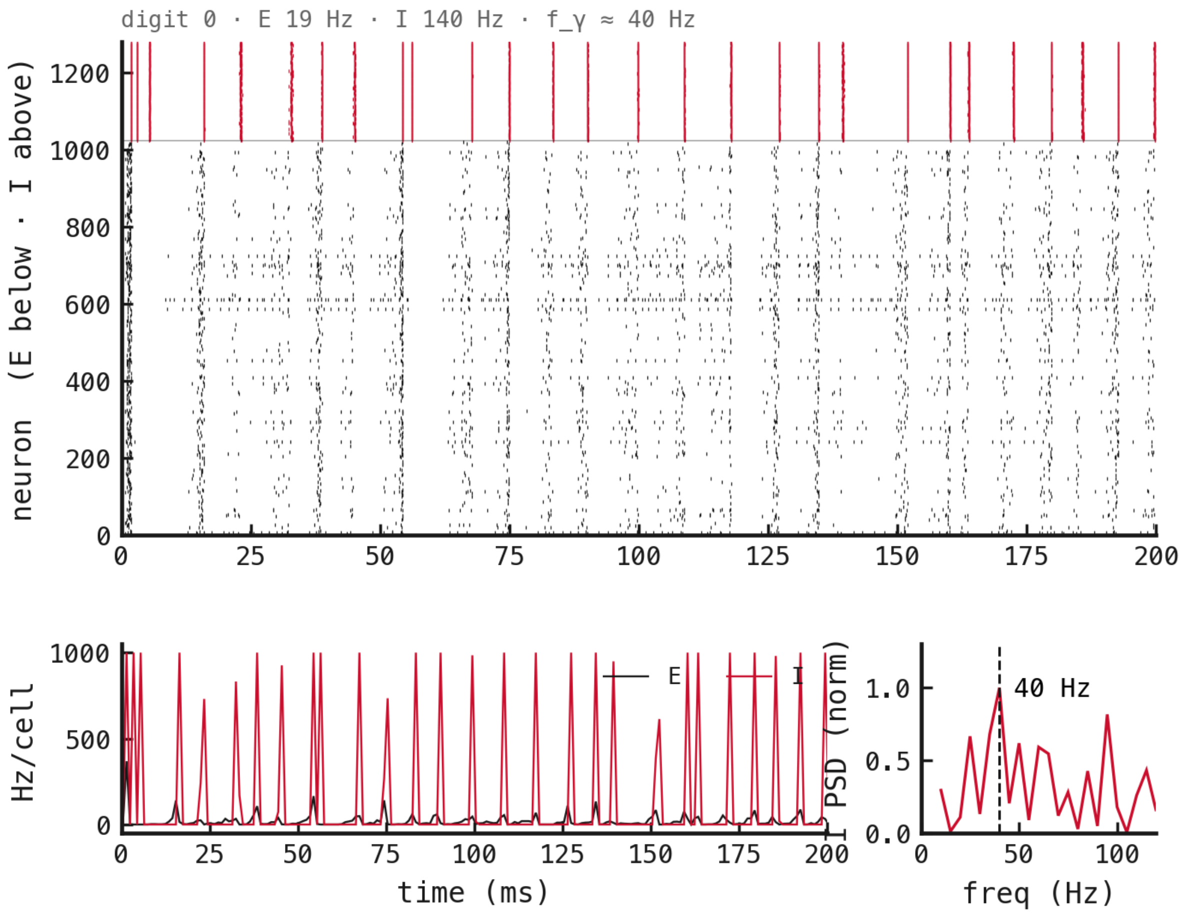

Figure 2: Sample raster: canonical PING, seed 42, held-out digit-0 probe.

TR-02 results — Spike-budget sweep

Run status: complete · 36/36 cells represented

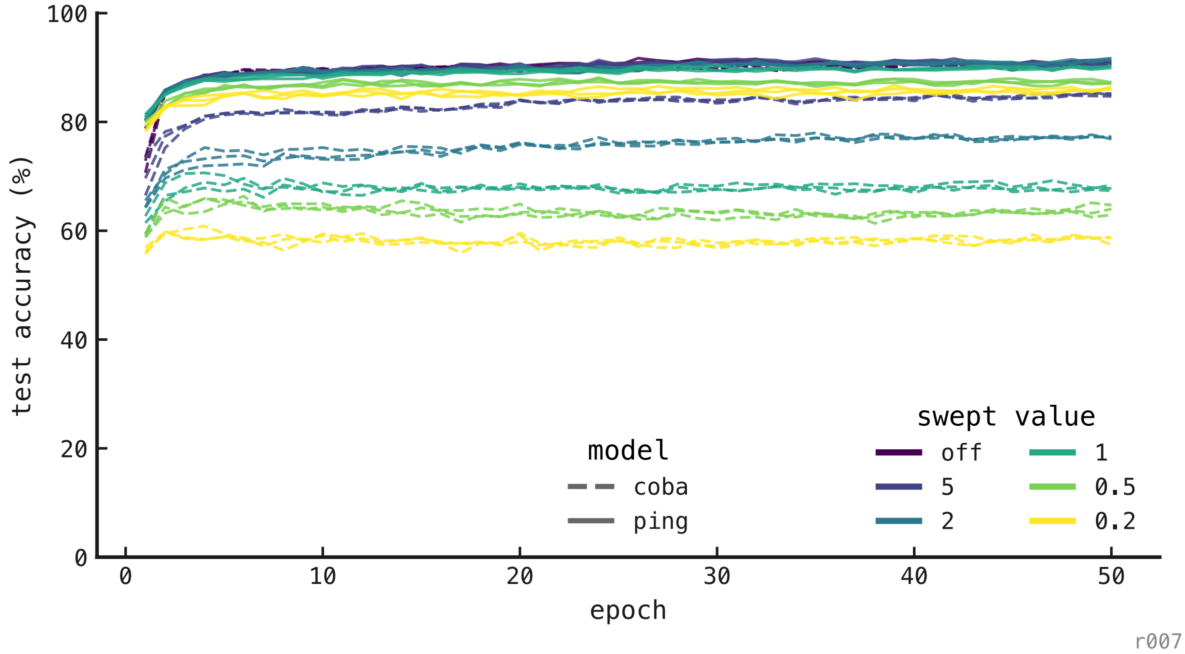

Figure 3: Training curves across both architectures, six spike budgets, and three seeds.

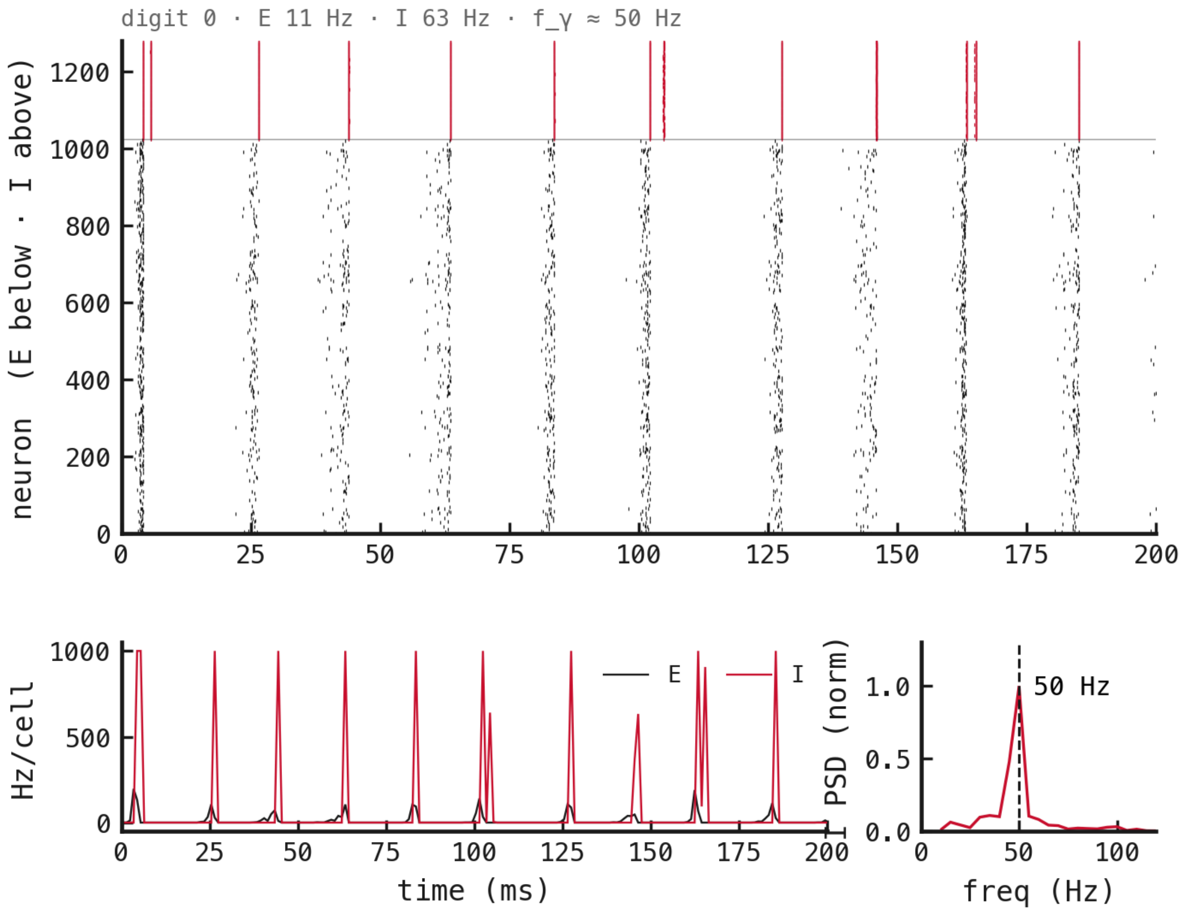

Figure 4: Sample raster: PING with spike budget off, seed 42.

TR-03 results — Inhibitory-timescale sweep

Run status: complete · 18/18 cells represented

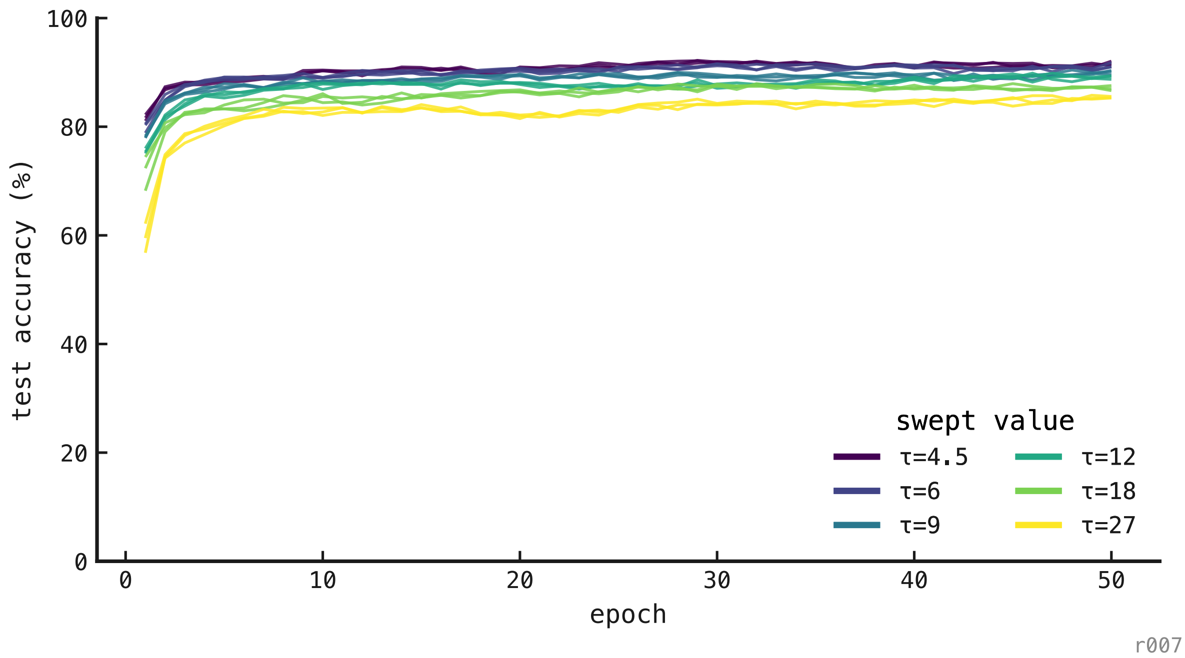

Figure 5: Training curves across six τ_{GABA} values and three seeds.

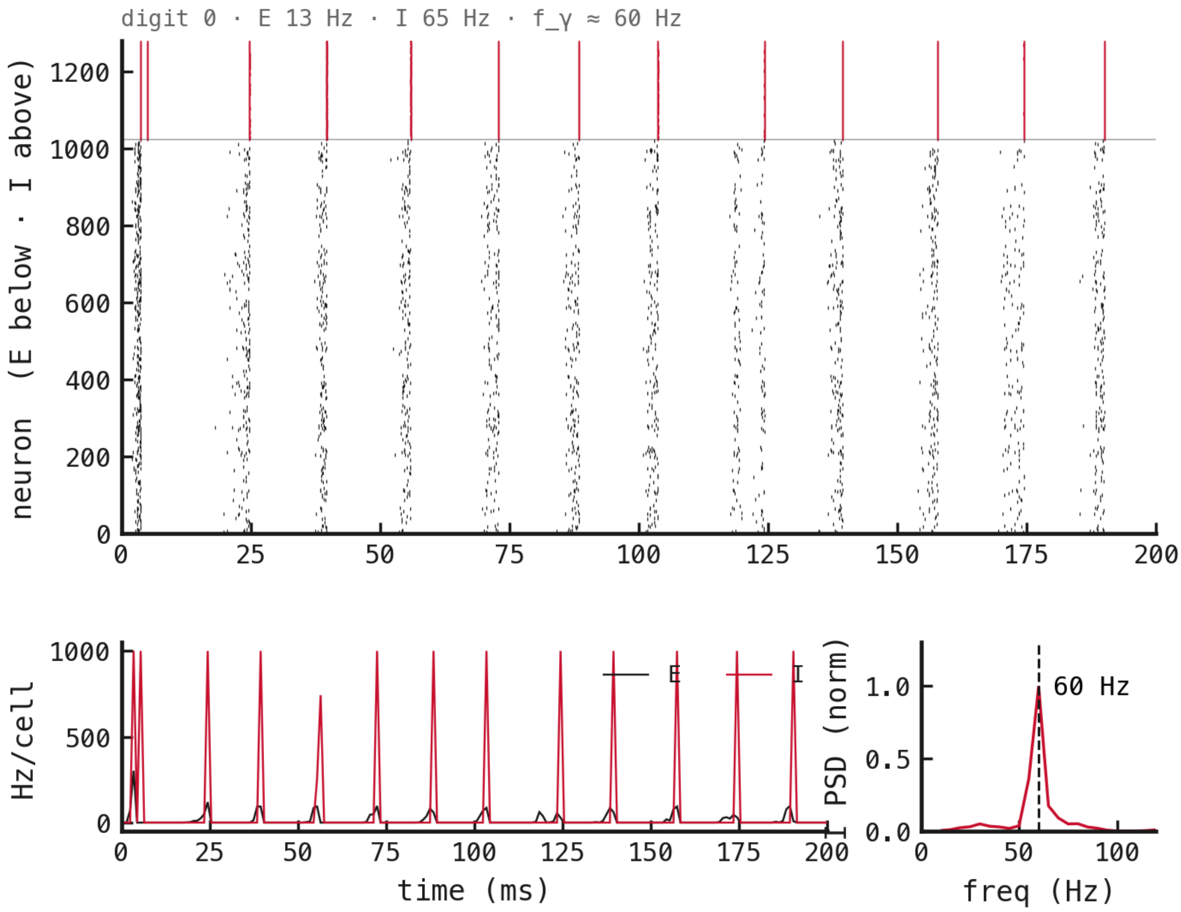

Figure 6: Sample raster: PING at $\tau_{\mathrm{GABA}} = 6$ ms, seed 42.

TR-04 results — Integration-timestep sweep

Run status: complete · 15/15 cells represented

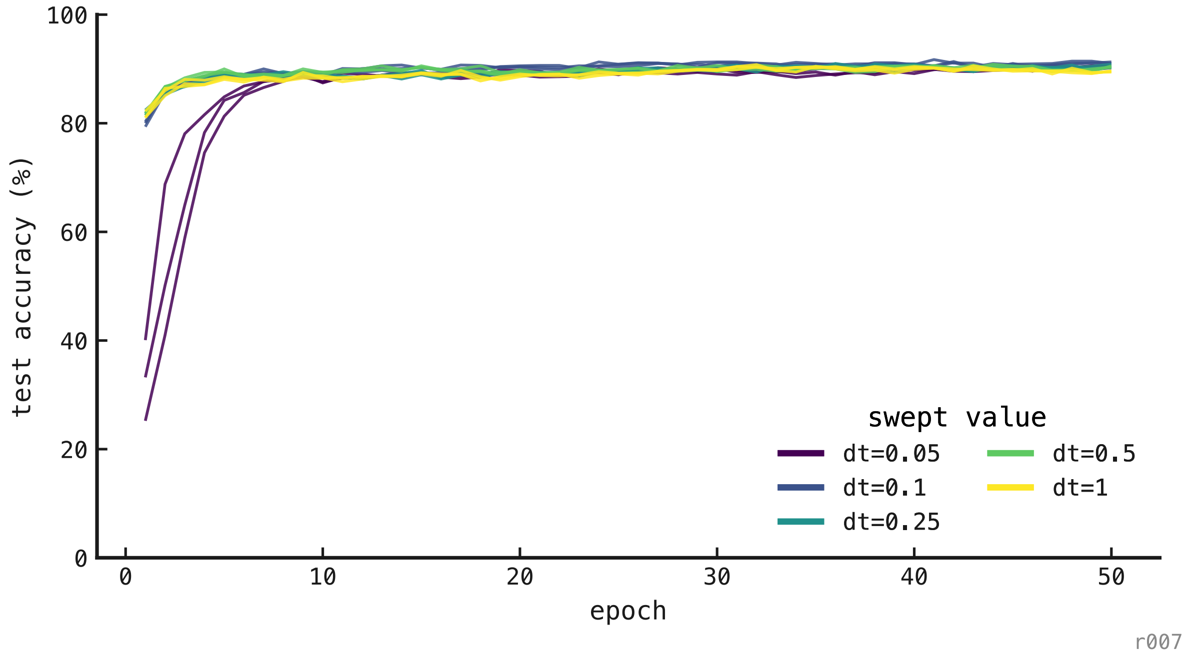

Figure 7: Training curves across five integration timesteps and three seeds.

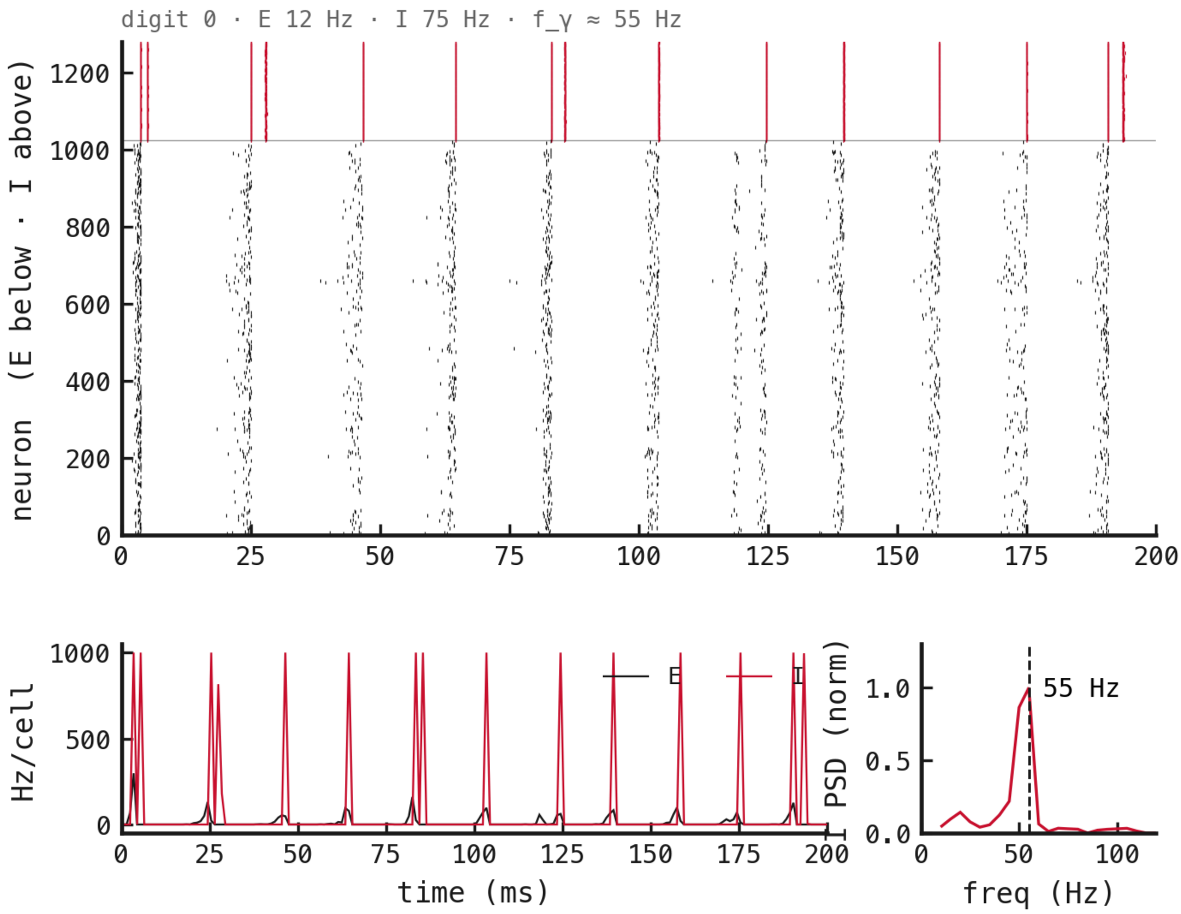

Figure 8: Sample raster: PING at $\Delta t = 0.1$ ms, seed 42.

TR-05 results — Recurrent-initialization sweep

Run status: complete · 12/12 cells represented

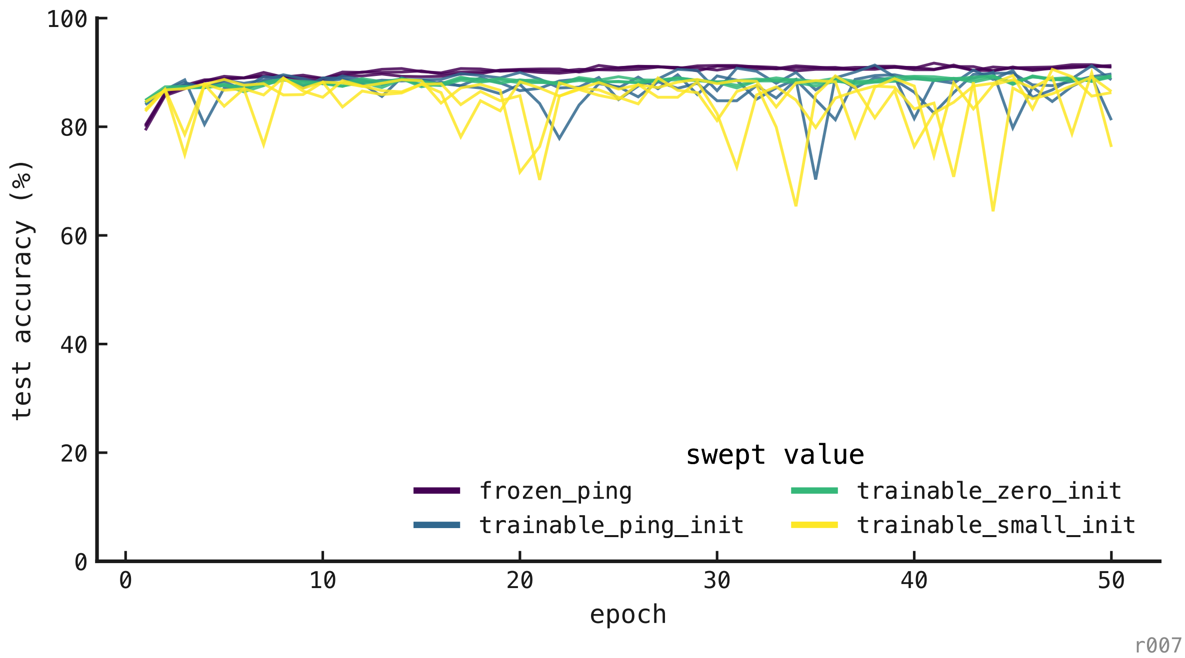

Figure 9: Training curves across four recurrent-loop conditions and three seeds.

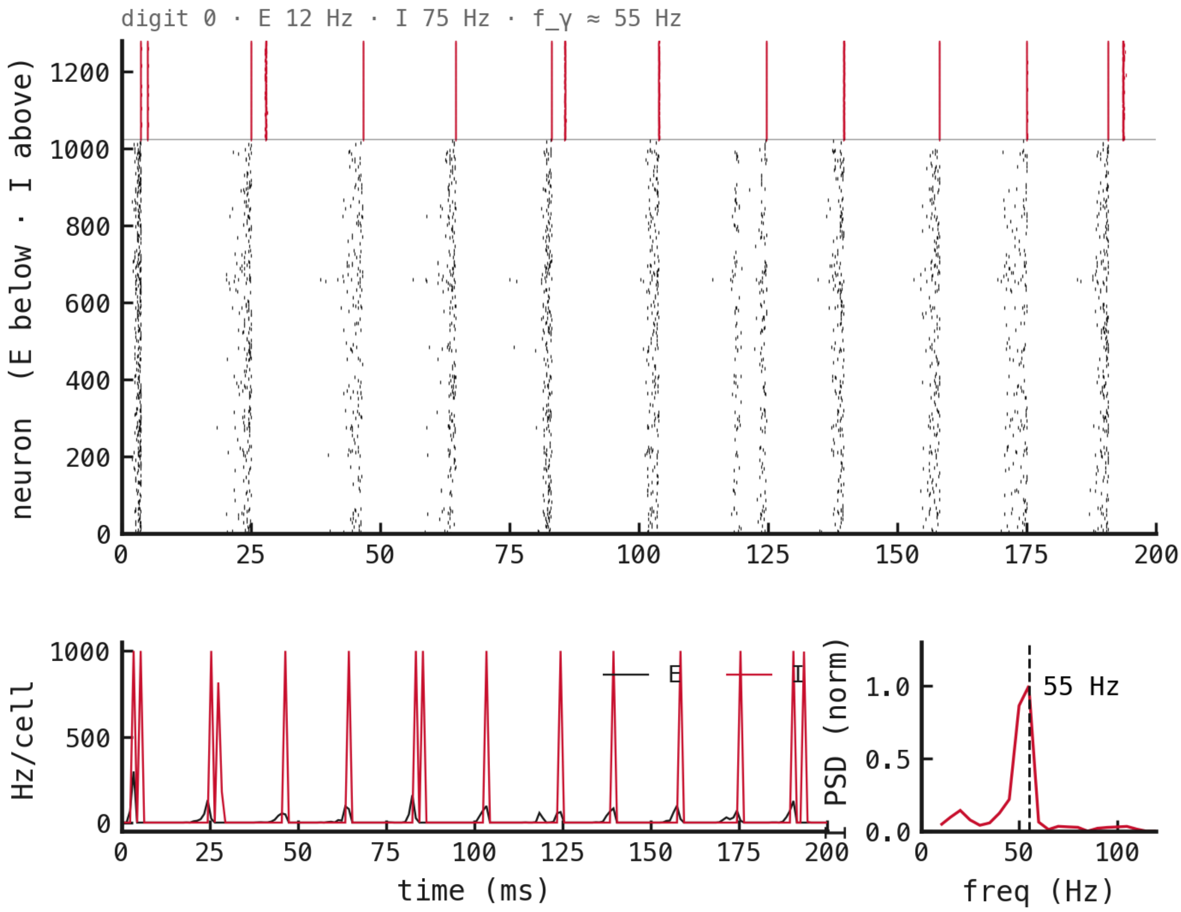

Figure 10: Sample raster: frozen PING recurrent loop, seed 42.

TR-06 results — Variable-rate streaming bank
Run status: pending · 0/3 cells trained

TODO — training curves. Add the three variable-rate learning curves after the Cambridge jobs complete.

TODO — sample raster. Add a seed-42 E/I/output raster at a declared held-out rate, plus low- and high-rate diagnostics if one raster hides a rate-dependent failure.